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Kongju National University, 1223-24 Cheonan-daero, Seobuk-gu, Cheonan-si, Chungcheongnam-do, Republic of Korea

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RESEARCH INTERESTS

Proton Exchange Membrane Fuel Cells (PEMFCs), Computational Fluid Dynamics (CFD), Physics-Informed Neural Networks (PINNs), Multi-physics Transport Modeling, Surrogate Modeling, Flow-field Design Optimization, AI-assisted Fuel Cell Systems, Digital Twin for Fuel Cell Systems

EDUCATIONAL BACKGROUND

PhD Candidate. in Mechanical Engineering

Kongju National University Cheonan Campus,

Sep 2026 — Present 2026

Cheonan City, Republic of Korea

Research Interest: Fuel Cell: PEMFCs, CFD, PIMLs, ML/DL, surrogate modeling

Supervisor: Professor Kim Hyun Chul

M.Sc. in Mechanical Engineering

Kongju National University Cheonan Campus,

Sep 2024 — Aug 2026

Cheonan City, Republic of Korea

Research Interest: PEMFCs, CFD, PINNs, surrogate modeling, flow-field design optimization

Cumulative GPA: 4.11/4.5

Thesis Title: AI-Assisted Multi-Objective Optimization of Morpho-Baffle Flow Fields in Proton Exchange Membrane Fuel Cells

Supervisor: Professor Kim Hyun Chul

B.Eng. in Mechanical Engineering

The University of Da Nang – University of Science and Technology,

Aug 2017 — May 2022

Da Nang City, Vietnam

Research Interest: Automotive engineering, electric vehicle design, fuel-saving vehicles, autonomous vehicle systems

Cumulative GPA: 6.9/10

Thesis Title: Designed an electric microbus concept and conducted basic calculations for powertrain, battery system, and vehicle layout

Supervisor: Dr. Nguyen Van Dong

AWARDS AND SCHOLARSHIPS

Graduate Researcher Stipend Kongju National University, Republic of Korea

Sep 2024 – Present

Research Assistant Scholarship Kongju National University, Republic of Korea

Sep 2024 – Present

Third Prize, Honda Eco Mileage Challenge. Ha Noi, Viet Nam

Jul 2018

Top 10 Asia Vehicle Design, Shell Eco-Marathon Asia. Sepang, Malaysia

May 2019

Full scholarship, Sakura Science Exchange Program. Yokohama National University, Japan

Nov 2019

Third Prize, Student Scientific Research Conference, University of Da Nang, Viet Nam

Dec 2021

RESEARCH EXPERIENCE

Kongju National University, Intelligent Manufacturing Technology and Fuel Cell Lab
Korea

Cheonan, Republic of

Graduate Researcher

Sep 2024 – Present

- Conducting research on proton exchange membrane fuel cells (PEMFCs), focusing on experimental performance analysis, CFD simulation, and AI-assisted modeling.
- Designed and analyzed PEMFC flow-field configurations, including conventional serpentine and baffle-enhanced serpentine channels.
- Performed single-cell PEMFC experiments under different temperature and pressure conditions to evaluate polarization behavior, power density, and operating performance.
- Conducted CFD simulations to investigate flow distribution, oxygen transport, pressure drop, and mass-transfer characteristics in PEMFC micro flow channels.

- Developed surrogate modeling and optimization frameworks for baffle geometry design to improve PEMFC power output and mass transport.
- Prepared research outputs for journal publications and conference presentations in the fields of PEMFC design, CFD, and machine learning.

Physics-Informed Neural Network Framework for PEMFC Transport Prediction Cheonan, Republic of Korea
Graduate Research Project 2025 – Present

- Developing a physics-informed neural network (PINN) framework to predict multi-physics transport fields in PEMFCs.
- Constructed CFD-based datasets, including boundary condition data, interface data, collocation points, anchor data, and validation data.
- Focused on predicting key transport variables such as velocity components, pressure, and oxygen mass fraction across channel, gas diffusion layer, and catalyst layer domains.
- Designed a PINN workflow including data preprocessing, normalization, model construction, physics-loss formulation, training, and validation against CFD results.
- Aimed to reduce the computational cost of high-fidelity CFD simulations and support rapid PEMFC design evaluation.

RESEARCH FUNDING

- **Basic Science Research Program, NRF–MSIT**, Graduate Researcher, Republic of Korea, 2024–Present. Research supported by the Basic Science Research Program funded by the Ministry of Science and ICT through the National Research Foundation of Korea (NRF), Grant No. 2021R1I1A3048752.
- **KETEP–MOTIE Funded Research Project**, Graduate Researcher, Republic of Korea, 2024–Present. Research supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry & Energy (MOTIE), Project No. RS-2024-00394769.
- **NRF–MSIT Funded Research Project**, Graduate Researcher, Republic of Korea, 2026–Present. Research supported by the National Research Foundation of Korea (NRF), funded by the Ministry of Science and ICT (MSIT), Project No. RS-2026-25487925.

PROFESSIONAL EXPERIENCE

DRÄXLMAIER Group

Technical Production Management Engineer / Technical Engineer TPM

Da Nang, Viet Nam

Jul 2022 — Aug 2024

- Worked in technical product management and production planning for automotive electrical systems.
- Prepared and managed master data lists, production plans, and technical documentation for modifications and rework.
- Created production process documentation, including machine setting data, work sequences, and operation instructions.
- Identified improvement potential in existing production plans and supported process improvement activities.
- Managed customer and product data in production documentation according to applicable standards and production-technical requirements.
- Coordinated with internal departments and international customers to support production, quality, and engineering-related tasks.

PROJECTS

Design of a 16-Seat Electric Bus for the Da Nang Eco City Project

Graduation Project

Dec 2022

- Designed an electric microbus concept and conducted basic calculations for powertrain, battery system, and vehicle layout.

Autonomous Golf Vehicle Research Project

University Scientific Research Project

2020 – 2022

- Developed a small-scale autonomous vehicle platform integrating Raspberry Pi, Arduino, camera modules, safety sensors, and deep-learning-based object recognition.
- Used Python, CARLA/Unreal Engine, CATIA, and AutoCAD for system design and simulation.

Design and Manufacturing of Fuel-Saving Vehicle

Honda Eco Mileage Challenge / Shell Eco-Marathon Asia

2017 – 2019

- Designed and manufactured high-efficiency fuel-saving vehicle prototypes.
- Worked on mechanical design, powertrain improvement, aerodynamic optimization, and Arduino-based control systems.
- Achieved Third Prize at the Honda Eco Mileage Challenge and Top 10 Asia Vehicle Design at Shell Eco-Marathon Asia.

PUBLICATIONS / MANUSCRIPTS

Published

- **Le Lit**, Thanh Duc Vuong, Hyun Chul Kim, “Multi-surrogate modeling for optimizing baffle geometry to improve power and mass transfer in micro flow channels of PEMFC,” *Fuel*, 2026, doi: 10.1016/j.fuel.2025.138240. (**IF: 7.5**)
- Pham Quoc Thai, Tran Thuan Hoang, Duong Van Hoa, Nong Trong Tu, Du Van Ngan, Ngo Tan Thong, Phan Van Binh, Huynh Duc Tri, and **Le Lit**, “Estimation of the state of lithium-ion battery for electric vehicles using Extended Kalman Filter,” *Lecture Notes in Networks and Systems*, vol. 1610, pp. 277–285, 2025 . doi: 10.1007/978-3-032-03856-2_30.

In Preparation

- **Le Lit**, Hyun Chul Kim, “Experimental and CFD analysis of baffled PEMFC flow fields under varying temperature and pressure,” manuscript in preparation.
- **Le Lit**, Hyun Chul Kim, “Physics-informed neural network-based prediction of multi-physics transport in proton exchange membrane fuel cells,” manuscript in preparation.

CONFERENCES / PRESENTATIONS

Poster presentation of the topic: “A Physics-Informed Neural Network Framework for Predicting Multi-physics Transport Fields in PEM Fuel Cells” (*International Conference on Precision Engineering and Sustainable Manufacturing, PRESM 2026*) Singapore, 28/06/2026 – 03/07/2026

Poster presentation of the topic: “AI-Driven Optimization of Morpho-Baffles in PEMFC Using CFD and Surrogate Modeling” (*KSMTE 2025 Autumn Conference*) Busan, Republic of Korea, 09/12/2025 – 12/12/2025

Oral presentation of the topic: “Surrogate Modeling for Optimizing Baffle Geometry to Improve Power and Mass Transfer in Micro Flow Channels of PEMFC” (*KSPE 2025 Autumn Conference*) Yeosu, Republic of Korea, 12/11/2025 – 14/11/2025

Oral presentation of the topic: “Morpho-Baffles: Adaptive Flow Field Design for PEMFC Using Machine Learning” (*2025 Spring Conference of the Korean Society of Hydrogen and New Energy*) Daegu, Republic of Korea, 14/05/2025 – 16/05/2025

PEER-REVIEWER ACTIVITIES

Energy Conversion and Management (IF: 10.9) – 1 paper
Under review

- **Paper focus:** PEMFC, thermal management, electrochemical performance, and cooling flow-field design May 2026

LANGUAGES

Test of English for International Communication (TOEIC)

- **Total score:** 740
- **Issuing organization:** ETS
- **Test date:** May 10, 2026 [Validity: 2 years]

CERTIFICATES AND TRAINING

Student Scientific Research Conference

- **Achievement:** Third Prize
- **Issuing organization:** The University of Da Nang
- **Issue date:** 2022

Shell Eco-Marathon Asia Certificates

- **Achievement:** High achievement and technical qualification certificates
- **Issue date:** 2019–2021

Sakura Science Exchange Program

- **Issuing organization:** Japan Science and Technology Agency, Yokohama National University
- **Issue date:** November 2019

Honda Eco Mileage Challenge Certificate

- **Achievement:** Third Prize, Vietnam Student Group
- **Issue date:** 2019

Assistant for Exchange Student Program

- **Issuing organization:** FPT University and Shibaura Institute of Technology
- **Issue date:** 2018

COMPUTER / TECHNICAL SKILLS

- **Programming:** Python, MATLAB, basic C/C++, LaTeX/Overleaf.
- **CFD and Simulation Programs:** ANSYS Fluent, PEMFC simulation.
- **Machine Learning and Data Analysis:** Physics-Informed Neural Networks, surrogate modeling, data preprocessing, normalization, model validation, regression analysis.
- **CAD and Design Programs:** CATIA, AutoCAD.
- **Experimental and Engineering Skills:** Single-cell PEMFC testing, polarization curve analysis, pressure drop analysis, uncertainty analysis, Arduino, Raspberry Pi, sensors.
- **Office and Documentation:** MS Word, Excel, PowerPoint, MS Project, technical reporting.

ACADEMIC REFERENCES

Name: Prof. Hyun Chul Kim

Position: Professor

Department: Department of Mechanical Engineering, Kongju National University

Email: khc@kongju.ac.kr

Name: Assoc. Prof. Pham Quoc Thai

Position: Associate Professor

Department: Department of Mechanical Engineering, The University of Da Nang – University of Science and Technology

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Name: Dr. Le Minh Tien

Position: Lecturer

Department: Department of Mechanical Engineering, The University of Da Nang – University of Science and Technology

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Name: Dr. Trinh Xuan Long

Position: Lecturer

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